ROBOTICS 4/M

TOPIC 7: JACOBIAN AND DIFFERENTIAL OPERATOR

7.1 DIFFERENTIAL MOTIONS OF A ROBOT

Previously the differential kinematics have been considered with respect to a moving reference frame. This reference frame could represent the hand of the robot and how it moves. However, this does not take into account the joints' differential motions that make the robot, and ultimately the hand, move. In order to find how the differential motions of the joints relate to the differential motion of the end effector, the configuration of the robot must be considered. This provides the information on how the joints move differentially. Once this is known, the relationship between the end effector and joint relative differential motions can be derived. This is derived by using *The Jacobian* [1,2].

7.2 THE JACOBIAN

The *Jacobian* is a representation of the geometry of the elements of a robot in time. It can be used to convert the differential motions/velocities of individual joints to differential motions/velocities of a particular point of interest (e.g. the end effector) [1,2]. As it defines the individual motions of joints, it can be used to relate these motions to the overall motion of the robot in question.

The Jacobian itself is a matrix relationship that is formed from the differentiation of the position relationships of a robot. If we define the position and orientation of the robot's end effector as the following set of equations:

$$Y_i = f_i\left(x_1, x_2, x_3 \cdots, x_j\right), \forall i \tag{7.1}$$

Here x represents the joint position and orientation. The differential relationships are obtained by differentiating (7.1) with respect to the differential changes in x_i i.e.

$$\delta Y_i = \frac{\partial f_i}{\partial x_1}\delta x_1 + \frac{\partial f_i}{\partial x_2}\delta x_2 + \frac{\partial f_i}{\partial x_3}\delta x_3 + \cdots + \frac{\partial f_i}{\partial x_j}\delta x_j, \forall i \tag{7.2}$$

The resulting set of differential equations can be written in the following matrix form:

$$\left[\delta Y_i\right] = \left[\frac{\partial f_i}{\partial x_j}\right]\left[\delta x_j\right]$$

$$\begin{bmatrix} \delta Y_1 \\ \delta Y_2 \\ \delta Y_3 \\ \vdots \\ \delta Y_i \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} & \cdots & \frac{\partial f_1}{\partial x_j} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} & \cdots & \frac{\partial f_2}{\partial x_j} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} & \cdots & \frac{\partial f_3}{\partial x_j} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_i}{\partial x_1} & \frac{\partial f_i}{\partial x_2} & \frac{\partial f_i}{\partial x_3} & \cdots & \frac{\partial f_i}{\partial x_j} \end{bmatrix} \cdot \begin{bmatrix} \delta x_1 \\ \delta x_2 \\ \delta x_3 \\ \vdots \\ \delta x_j \end{bmatrix} \tag{7.3}$$

The i by j matrix is called the *Jacobian*. For a particular robot manipulator the relationship of the differential kinematics of the end effector reference frame to the joint differential motions can be formed using the Jacobian [1,2] i.e.

$$[D]=[J][D_\theta]$$

$$\begin{bmatrix} dx \\ dy \\ dz \\ \delta x \\ \delta y \\ \delta z \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial \theta_1} & \frac{\partial f_1}{\partial \theta_2} & \frac{\partial f_1}{\partial \theta_3} & \frac{\partial f_1}{\partial \theta_4} & \frac{\partial f_1}{\partial \theta_5} & \frac{\partial f_1}{\partial \theta_6} \\ \frac{\partial f_2}{\partial \theta_1} & \frac{\partial f_2}{\partial \theta_2} & \frac{\partial f_2}{\partial \theta_3} & \frac{\partial f_2}{\partial \theta_4} & \frac{\partial f_2}{\partial \theta_5} & \frac{\partial f_2}{\partial \theta_6} \\ \frac{\partial f_3}{\partial \theta_1} & \frac{\partial f_3}{\partial \theta_2} & \frac{\partial f_3}{\partial \theta_3} & \frac{\partial f_3}{\partial \theta_4} & \frac{\partial f_3}{\partial \theta_5} & \frac{\partial f_3}{\partial \theta_6} \\ \frac{\partial f_4}{\partial \theta_1} & \frac{\partial f_4}{\partial \theta_2} & \frac{\partial f_4}{\partial \theta_3} & \frac{\partial f_4}{\partial \theta_4} & \frac{\partial f_4}{\partial \theta_5} & \frac{\partial f_4}{\partial \theta_6} \\ \frac{\partial f_5}{\partial \theta_1} & \frac{\partial f_5}{\partial \theta_2} & \frac{\partial f_5}{\partial \theta_3} & \frac{\partial f_5}{\partial \theta_4} & \frac{\partial f_5}{\partial \theta_5} & \frac{\partial f_5}{\partial \theta_6} \\ \frac{\partial f_6}{\partial \theta_1} & \frac{\partial f_6}{\partial \theta_2} & \frac{\partial f_6}{\partial \theta_3} & \frac{\partial f_6}{\partial \theta_4} & \frac{\partial f_6}{\partial \theta_5} & \frac{\partial f_6}{\partial \theta_6} \end{bmatrix} \cdot \begin{bmatrix} d\theta_1 \\ d\theta_2 \\ d\theta_3 \\ d\theta_4 \\ d\theta_5 \\ d\theta_6 \end{bmatrix} \tag{7.4}$$

Here $[J]$ is the *robot Jacobian* [1,2], $[D]$ is the differential position and orientation of the end effector and $[D_\theta]$ is the differential joint motions. In $[D]$, dx, dy and dz are the differential motions along the x, y and z axes, and δx, δy and δz are the differential rotations about the x, y and z axes. In $[D_\theta]$, each $d\theta$ represents the motion of the particular joints of the robot. If these two differential motions are divided by dt then (7.4) is transformed from the differential motion matrix to a velocity matrix.

CALCULATING THE JACOBIAN

The elements of the Jacobian are the derivatives of the corresponding kinematic equations with respect to one of the joint variables [1,2]. In equation (7.4) the first element of $[D]$ is dx, which indicates that the first kinematic equation must represent movements along the x-axis (i.e. motion relative to p_x). It follows that the derivative of p_x is dx. The same argument holds for dy and dz.

However, since there are only components of the orientation vectors about that axes, there is no unique equation defining the rotations about the axes. Therefore there is no single equation available for differential rotations about the axes.

THE JACOBIAN RELATIVE TO THE LAST FRAME

It has been found that it is simpler to calculate the Jacobian Relative to the last frame T_6. Therefore the velocity relationship can be written thus [3,1]:

$$\left[{}^{T6}D\right]=\left[{}^{T6}J\right]\left[D_\theta\right] \tag{7.5}$$

This means that for the same joint's differential motions, premultiplied with the Jacobian relative to the last frame, the end effector's differential motions will be given relative to the last frame. To calculate the Jacobian relative to the last frame using the following relationships [3]:

- The differential kinematics relationship can be written as

$$\begin{bmatrix} {}^{T6}dx \\ {}^{T6}dy \\ {}^{T6}dz \\ {}^{T6}\delta x \\ {}^{T6}\delta y \\ {}^{T6}\delta z \end{bmatrix} = \begin{bmatrix} {}^{T6}J_{11} & {}^{T6}J_{12} & \cdots & {}^{T6}J_{16} \\ {}^{T6}J_{21} & {}^{T6}J_{22} & \cdots & {}^{T6}J_{26} \\ {}^{T6}J_{31} & {}^{T6}J_{32} & \cdots & {}^{T6}J_{36} \\ {}^{T6}J_{41} & {}^{T6}J_{42} & \cdots & {}^{T6}J_{46} \\ {}^{T6}J_{51} & {}^{T6}J_{52} & \cdots & {}^{T6}J_{56} \\ {}^{T6}J_{61} & {}^{T6}J_{62} & \cdots & {}^{T6}J_{66} \end{bmatrix} \cdot \begin{bmatrix} d\theta_1 \\ d\theta_2 \\ d\theta_3 \\ d\theta_4 \\ d\theta_5 \\ d\theta_6 \end{bmatrix} \tag{7.6}$$

- Assuming that any combination of transformation matrices $A_1A_2...A_n$ can be expressed as an appropriate n,o,a,p matrix. The corresponding element in this matrix are used to define the Jacobian elements in Equation (7.6).
- If joint i is a revolute joint, then:

$$\begin{matrix} {}^{T6}J_{1i}=\left(-n_x p_y + n_y p_x\right) & {}^{T6}J_{2i}=\left(-o_x p_y + o_y p_x\right) & {}^{T6}J_{3i}=\left(-a_x p_y + a_y p_x\right) \\ {}^{T6}J_{4i}=n_z & {}^{T6}J_{5i}=o_z & {}^{T6}J_{6i}=a_z \end{matrix} \tag{7.7}$$

- If joint I is a prismatic joint, then:

$$\begin{matrix} {}^{T6}J_{1i}=n_z & {}^{T6}J_{2i}=o_z & {}^{T6}J_{3i}=a_z \\ {}^{T6}J_{4i}=0 & {}^{T6}J_{5i}=0 & {}^{T6}J_{6i}=0 \end{matrix} \tag{7.8}$$

- The *n,o,a,p* components in (7.7) and (7.8) for column *I* are associated with the elements in ${}^{i-1}T_6$:

Column 1, use ${}^{0}T_6 = A_1A_2A_3A_4A_5A_6$

Column 2, use ${}^{1}T_6 = A_2A_3A_4A_5\,A_6$

Column 3, use ${}^{2}T_6 = A_3A_4A_5A_6$

Column 4, use ${}^{3}T_6 = A_4A_5A_6$

Column 5, use ${}^{4}T_6 = A_5A_6$

Column 6, use ${}^{5}T_6 = A_6$

Where *A* represents the respective transformation matrix for the joints concerned [3,1].

7.3 THE JACOBIAN AND DIFFERENTIAL OPERATOR

The Jacobian and the Differential Operator can be related together in the following way. From Equation (7.4) and knowing the Jacobian, the [*D*] matrix can be calculated (which contains the differential kinematic variables for the end effector). The elements of [*D*] can be substituted into the differential operator equation [1,2] i.e.

$$\Delta = \begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\delta z & \delta y & 0 \\ \delta z & 1 & -\delta x & 0 \\ -\delta y & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{7.9}$$

This differential operator can then be used to locate the new position and orientation of the end effectors frame. Alternatively, the [${}^{T6}D$] elements are for a moving reference frame and can therefore be used in the following equation to define the differential operator:

$$\left[T^{-1}\right]\left[\Delta\right]\left[T\right] = {}^{T6}\Delta = \begin{bmatrix} 0 & -{}^{T6}\delta z & {}^{T6}\delta y & {}^{T6}dx \\ {}^{T6}\delta z & 0 & -{}^{T6}\delta x & {}^{T6}dy \\ -{}^{T6}\delta y & {}^{T6}\delta x & 0 & {}^{T6}dz \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{7.10}$$

7.4 INVERSE JACOBIAN

To calculate the differential motions/velocities of the robot's joints required for desired differential motions/velocities of the end effector, the *Inverse Jacobian* must be derived and used in the following equations [1]:

$$\begin{gathered} \left[J^{-1}\right]\left[D\right] = \left[J^{-1}\right]\left[J\right]\left[D_\theta\right] \\ \therefore \left[D_\theta\right] = \left[J^{-1}\right]\left[D\right] \end{gathered} \tag{7.11}$$

Alternatively:

$$\begin{gathered} \left[{}^{T6}J^{-1}\right]\left[{}^{T6}D\right] = \left[{}^{T6}J^{-1}\right]\left[{}^{T6}J\right]\left[D_\theta\right] \\ \therefore \left[D_\theta\right] = \left[{}^{T6}J^{-1}\right]\left[{}^{T6}D\right] \end{gathered} \tag{7.12}$$

Therefore, if the Inverse Jacobian is known then the differential joint variables can be calculated. Calculating the inverse of the Jacobian can be a very complex undertaking since the elements of the Jacobian change with changes in the robots operational configuration. More detail of this can be found in [1].

REFERENCES

[1] Niku, S.B., *Introduction to Robotics*, Pearson Education, 2001.

[2] Sciavicco, L. & Siciliano, B., *Modelling and Control of Robot Manipulators*, Springer, 2001.

[3] Paul, R.P., *Robot Manipulators, Mathematics, Programming and Control*, The MIT Press, 1981.